Data Science and Artificial Intelligence

Machine Learning

Nearest Neighbour

Lecture No. 4

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Recap of Previous Lecture

- Topic KNN
- Topic
- Topic
- Topic
- Topic

Topics to be Covered

- Topic — KNN → time & space Complexity
- Topic — Decision tree
- Topic
- Topic
- Topic

"Never be limited by other people's limited imaginations."
Dr. Mae Jemison
(First female African-American astronaut)
Forbes INDIA

Revision * KNN → Questions

4. We have data from a survey and objective testing with two attributes A and B to classify whether a special paper tissue is good or not. Here are four training samples given in the table. Now the factory produces a new paper tissue that pass laboratory test with A = 3 and B = 7. If K = 3, then 'Good' is the classification of this new tissue?

True

A	B	C = Classification
7	6	Bad
7	4	Bad
4	4	Good
2	4	Good

a) True
b) False

Nearest Neighbour Method

Time Complexity in KNN

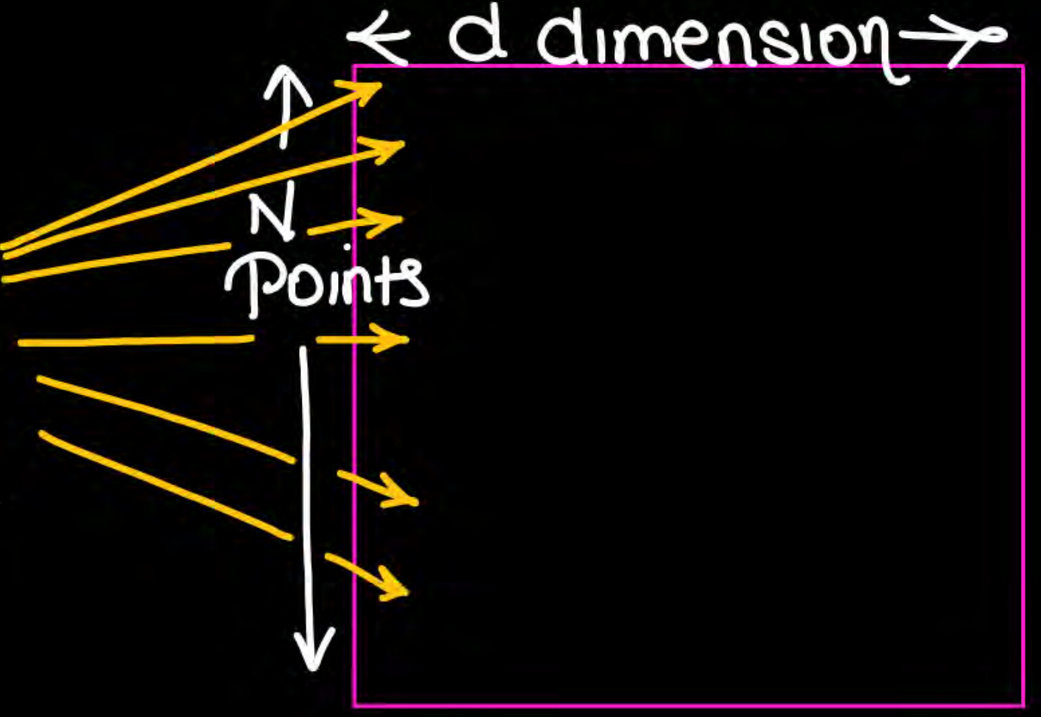
Training X

Testing ⇒

Any Testing point

for single distance

$$\left(\sum_{j=1}^{D}\left(x_T^j - x_i^j\right)^2\right)^{1/2}$$



- For single distance we need Computation which are order of $O(D)$
- For each test point we need to find distance from all N points

$\hookrightarrow O(ND)$

- we can see that Computations $\propto N$

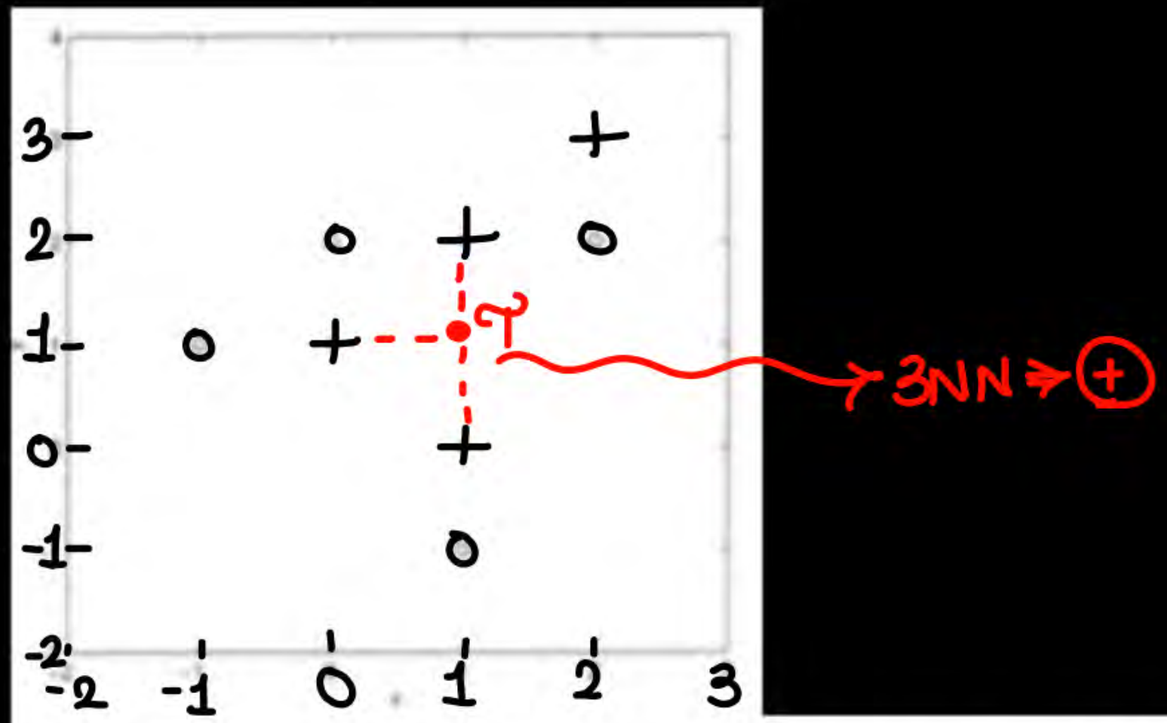
* The KNN algorithm is computationally extensive becoz $O(ND)$ calculation has to be done for each test point

Nearest Neighbour Method

Space Complexity in KNN

- In all previous algorithm after finding β we can delete the training data
- But here whole training data has to be stored in the memory to used @ time of testing

N points
D dimension $\Rightarrow O(ND)$



11) Suppose, you want to predict the class of new data point x=1 and y=1 using eucludian distance in 3-NN. In which class this data point belong to?

7. In a K-NN classification problem with K set to 5, if the distances of the five nearest neighbors from a data point are 1.5, 1.2, 1.8, 2.2, and 1.6, what is the predicted class using weighted voting with the inverse distance weighting method?

a. Class X

b. Class Y

c. Class Z

d. Insufficient information

1	2	3	4	5
↓	↓	↓	↓	↓
X	Y	Z	Z	Y

given in Question

$$\frac{\frac{X}{1.5}+\frac{Y}{1.2}+\frac{Z}{1.8}+\frac{Z}{2.2}+\frac{Y}{1.6}}{C \Rightarrow \sum \frac{1}{1.5}+\frac{1}{1.2}+\frac{1}{1.8}+\frac{1}{2.2}+\frac{1}{1.6}}$$

in Class

$$\Rightarrow Y \Rightarrow \frac{\sum y_i w_i}{\sum w_i}$$

Regression

So $\hat{Y} = \frac{1}{c}(.66x + 1.45Y + 1.012)$

$\hat{Y} = Y$ max Coef wala.

What is the main drawback of the KNN algorithm?

- [ ] Prone to overfitting
- [ ] Computationally expensive for large datasets
- [ ] Only applicable to linearly separable data
- [ ] Requires a large amount of training data

What is the impact of the "distance-weighted" voting scheme in KNN on the influence of neighbors?

- [ ] All neighbors contribute equally
- [x] Closer neighbors have more influence
- [x] Farther neighbors have ~~more~~ less influence
- [ ] It has no effect on neighbor influence

- weights ⇒ $1/d$, $1/d^2$
- $\hat{y} = \frac{\sum y_i w_i}{\sum w_i}$

Question 1: What is the KNN algorithm?

(A) The KNN algorithm is non-parametric and does not make assumptions about the underlying distribution of the data.

(B) The KNN works by finding the K closest data points (neighbors) to the query point and predicts the output based on the labels of these neighbors.

(C) The KNN algorithm is a lazy machine learning algorithm for classification and regression tasks. It can work well with both binary and multi-class classification problems.

(D) All of the above

D

Question 3: What are the disadvantages of using the KNN algorithm?

(A) As the number of dimensions increases, the distance between any two points in the space becomes increasingly large, making it difficult to find meaningful nearest neighbors.

(B) Computationally expensive, especially for large datasets, and requires a large amount of memory to store the entire dataset.

(C) Sensitive to the choice of K and distance metric.

(D) All of the above

D

Question 4: How do you choose the value of K (the number of neighbors to consider) in the KNN algorithm? (Select two)

(A) A small value of K, for example, K=1, will result in a more flexible model but may be prone to overfitting.

A,B

(B) A large value of K, for example, K=n, where n is the size of the dataset, will result in a more stable model but may not capture the local variations in the data.

(C) A large value of K, for example, K=n, where n is the size of the dataset, will result in a more flexible model but may be prone to overfitting.

(D) A small value of K, for example, K=1, will result in a more stable model but may not capture the local variations in the data.

Question 7: What are the ideal use cases for KNN?

(A) KNN is best suited for small to medium-sized datasets with relatively low dimensionality. It can be useful in situations where the decision boundary is linear. It can be effective in cases where the data is clustered or has distinct groups.

(B) KNN is best suited for large datasets with relatively high dimensionality. It can be useful when the decision boundary is highly irregular or nonlinear. It can be effective in cases where the data is clustered or has distinct groups.

(C) KNN is best suited for small to medium-sized datasets with relatively low dimensionality. It can be useful when the decision boundary is highly irregular or nonlinear. It can be effective in cases where the data is clustered or has distinct groups.

(D) KNN is best suited for small to medium-sized datasets with relatively low dimensionality. It can be useful when the decision boundary is highly irregular or nonlinear. It can be effective in cases where the data is not clustered or doesn't have distinct groups.

C

Question 5: How do you handle imbalanced data in the KNN algorithm?

(A) Weighted voting, where the vote of each neighbor is weighted by its inverse distance to the query point. This gives more weight to the closer neighbors and less weight to the farther neighbors, which can help to reduce the effect of the majority class.

(B) Oversample the minority class.

(C) Undersample the majority class.

(D) All of the above.

D

C

Question 8: How does the KNN algorithm work? (Select two)

(A) KNN works by calculating the distance between a data point and all other points in the dataset. Then, KNN selects the k-nearest neighbors. For regression, the most common class among the 'k' neighbors is assigned as the predicted class for the new data point.

(B) KNN works by calculating the distance between a data point and all other points in the dataset. Then, KNN selects the k-nearest neighbors. For classification, averages the values of the most common class among the 'k' neighbor to the target data point.

(C) KNN works by calculating the distance between a data point and all other points in the dataset. Then, KNN selects the k-nearest neighbors. For classification, the most common class among the 'k' neighbors is assigned as the predicted class for the new data point.

(D) KNN works by calculating the distance between a data point and all other points in the dataset. Then, KNN selects the k-nearest neighbors. For regression tasks, instead of a majority vote, the algorithm takes the average of the 'k' nearest neighbors' values as the prediction.

KNN

Decision Tree

Important for GATE exam

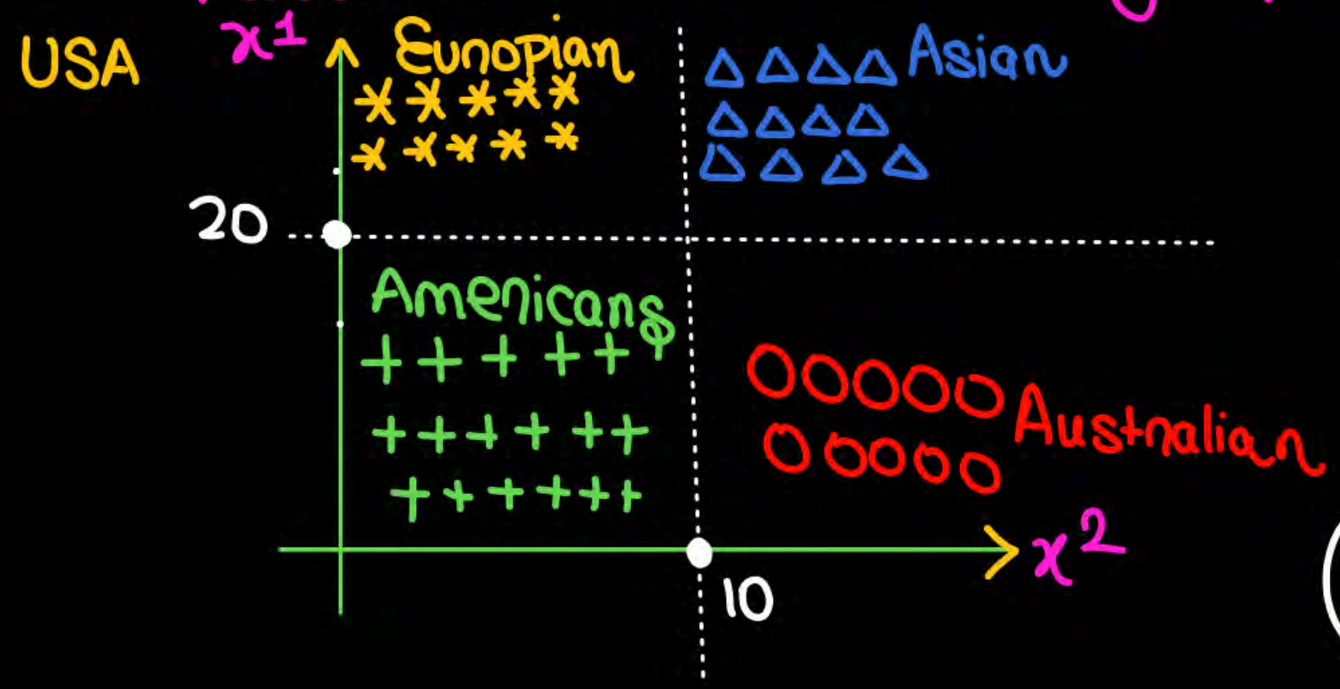
Numerical and theoretical Questions can be asked

Decision Tree

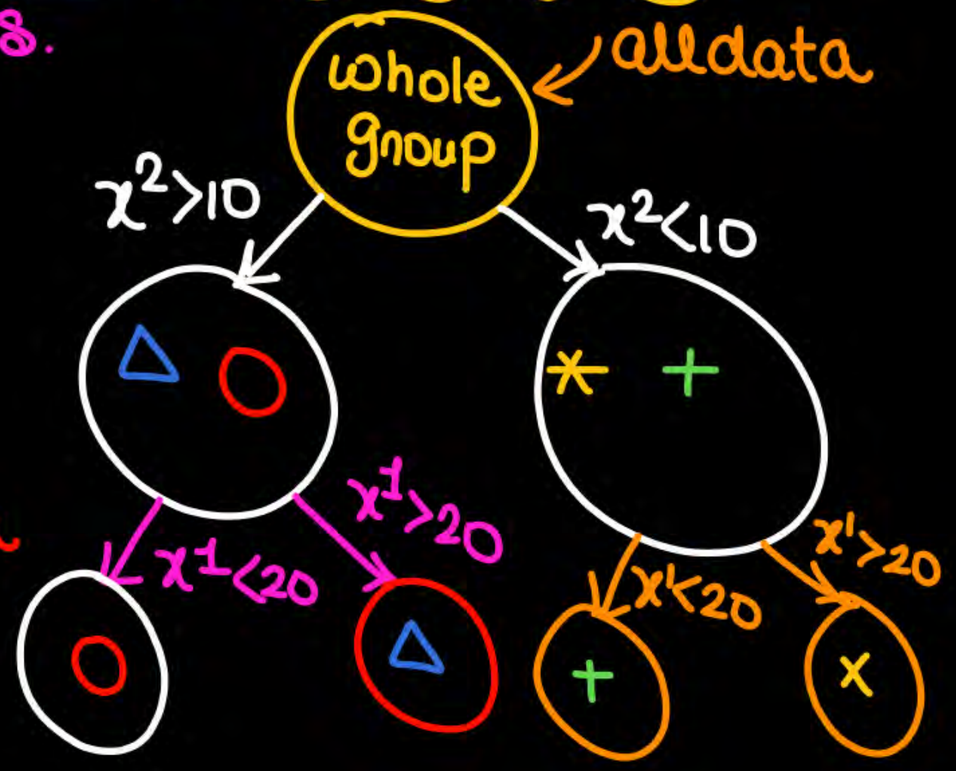
What is a Decision Tree

❖ How does it work lets see an example of 2D data.

↳ we divide data into ~~~~ groups.



• non parametric



What is a Decision Tree

❖ How does it work lets see an example of 2D data.

- So in a decisiontree ⇒ we start with whole and then keep on bifurcating/break on creating groups based on dimensions
- motive ⇒ we want to break the data into groups such that final groups are having homogeneous data.

Decision Tree

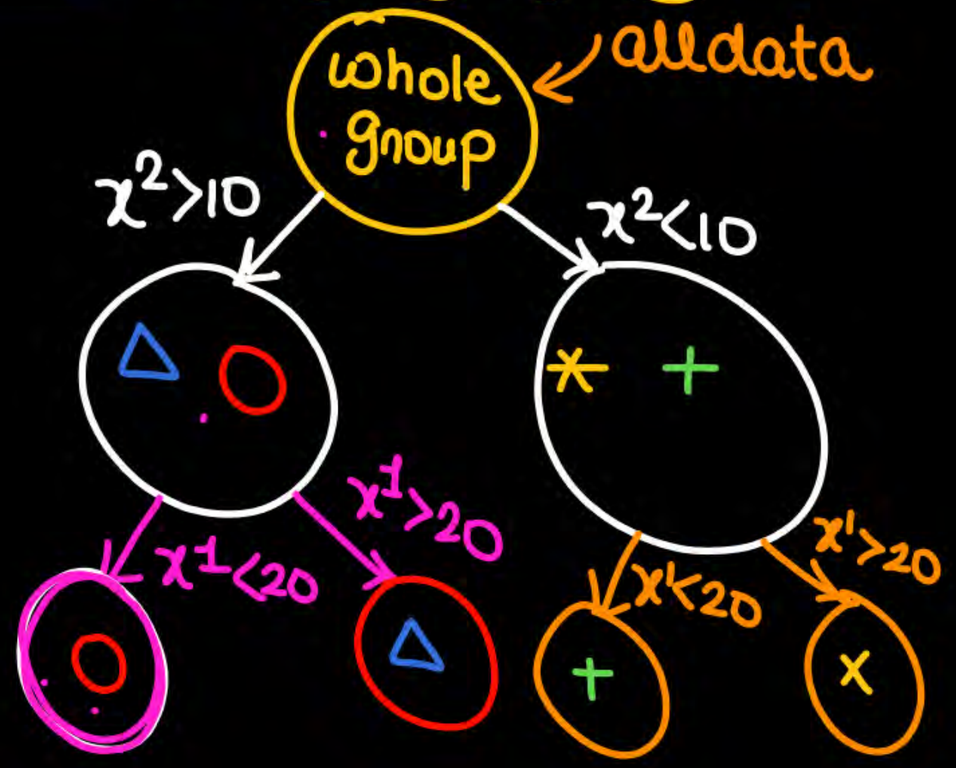
What is a Decision Tree

❖ How does it work lets see an example of 2D data.

So any new points ⇒

$X^2 = 30, X^1 = 5 \rightarrow$ Red

• non parametric

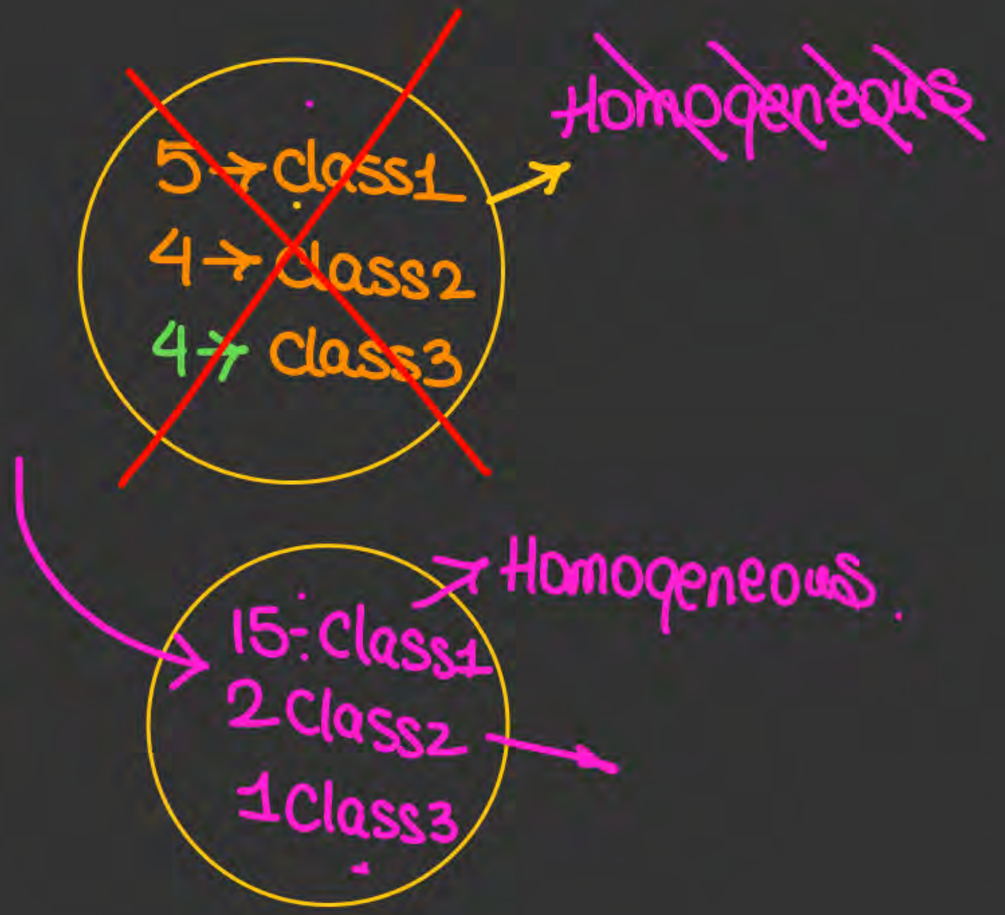


Why we need homogeneous data @ end nodes

↓

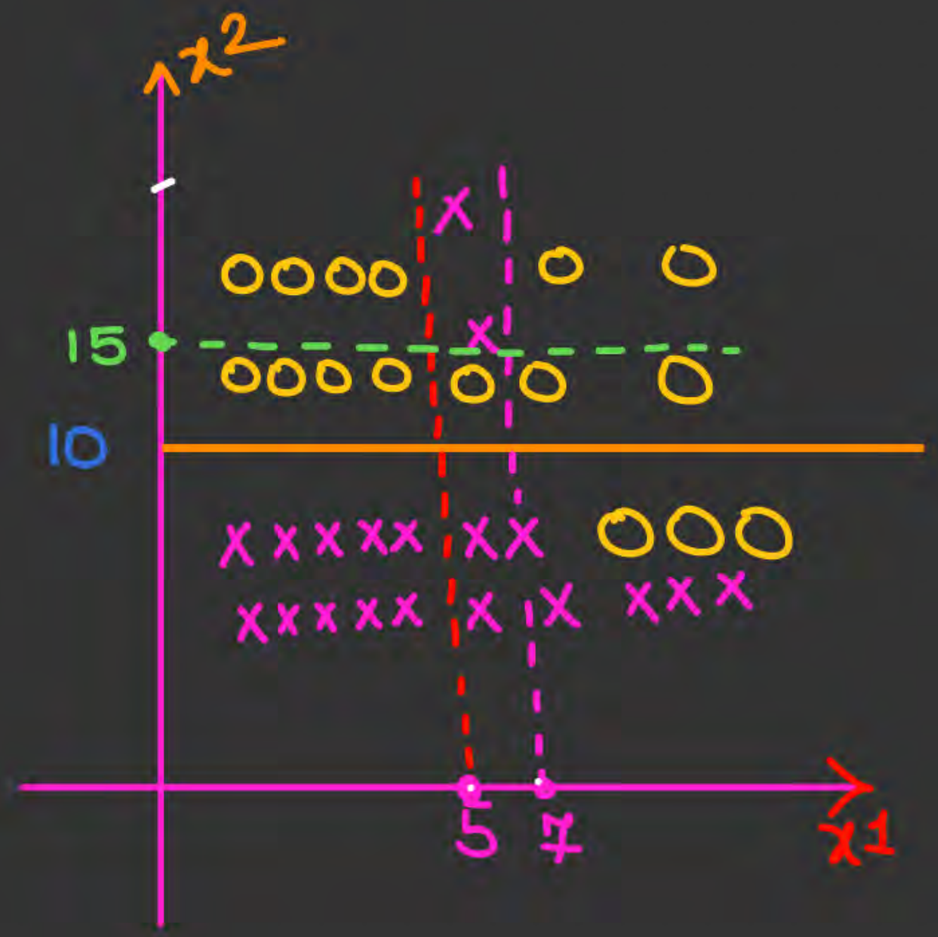
leaf nodes

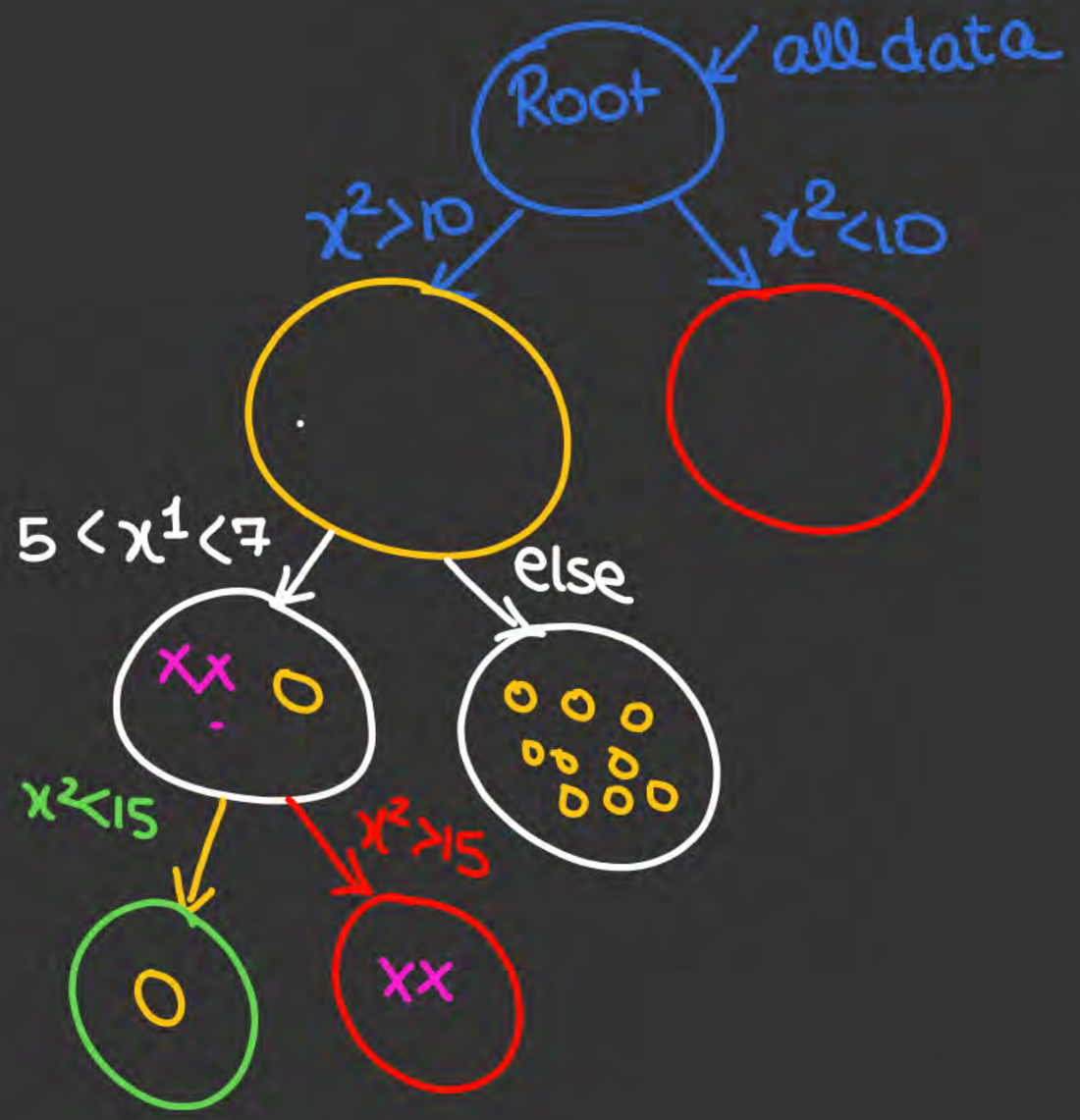
- So that the decision on $\hat{y}$ can be taken without confusion
-



If we have classification Problem then @ leaf node ⇒

- y_p = majority of class of points in leaf
- In Regression y_p = avg of y of all points in leaf


x^2
15
10
5
7
x^1


Root
all data
$x^2>10$
$x^2<10$
$5<x^1<7$
else
$x^2<15$
$x^2>15$

Decision Tree

What is a Decision Tree

So the target is ...

❖ How does it work lets see an example of 2D data.

- So the target is to get the homogeneous nodes, but it lead to overfitting on data
- So we have to remove overfitting and also how to decide when and where the data has to be bifurcated.

Decision Tree

What is a Decision Tree

So what is the algo..

- How does it work lets see an example of 2D data.

done

Decision Tree

What is a Decision Tree

- How does it work lets see an example of 2D data.

Why we bifurcate...

To reduce confusion

Decision Tree

What is a Decision Tree

- How does it work lets see an example of 2D data.

done

Decision tree is prone to overfit

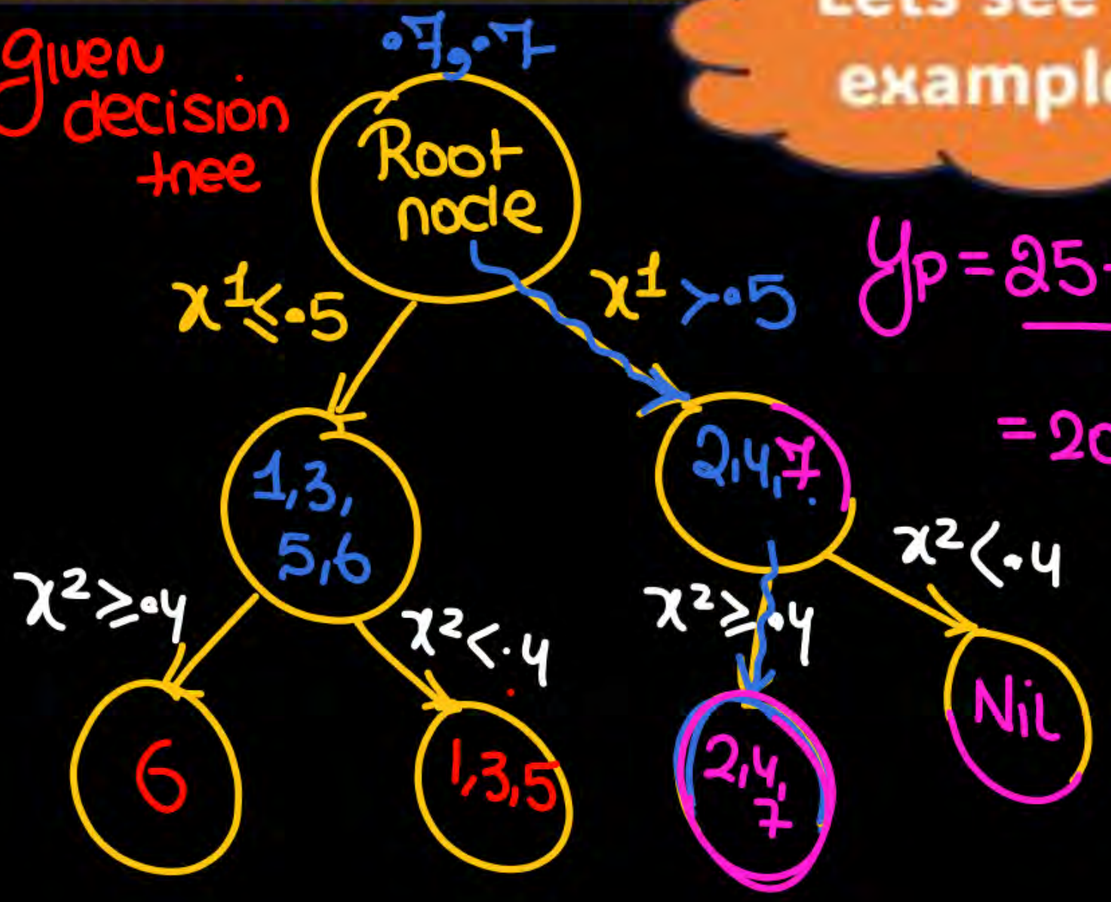
Decision Tree

Lets understand using an example

find $y_p \Rightarrow$ for (.7, .7)

	X1	X2	Y
1	0.5	0.3	15
2	0.8	0.6	25
3	0.4	0.3	12
4	0.7	0.4	18
5	0.3	0.1	6
6	0.2	0.5	9
7	0.9	0.7	18

given decision tree



Lets see an example..

$$y_p = \frac{25+18+18}{3} = 20.33.$$

Decision Tree

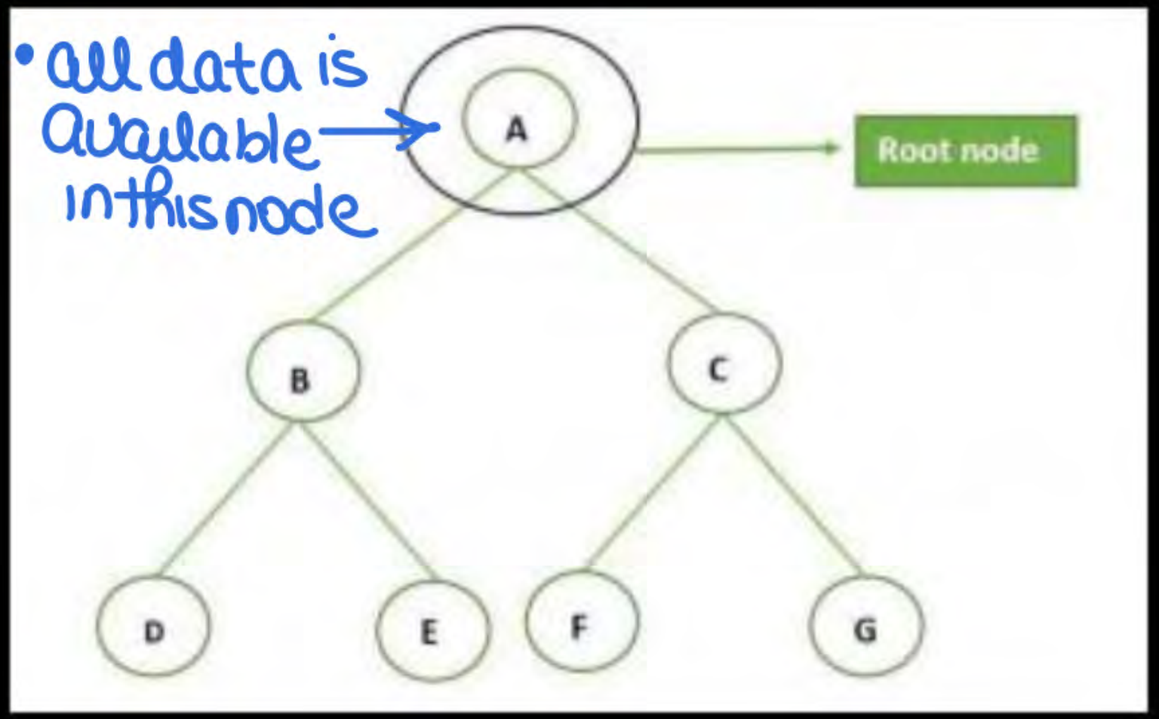
What is a Decision Tree

Some Important Terms

What is Root Node :-
A node that is the first or topmost node in a tree is called a root node. In every tree, there is always one root node, which is the only node that has never previously been connected to another node.

- Decision/Internal node are those which have children.
- Leaf Node are those which donot have any children.

• all data is available in this node



Decision Tree

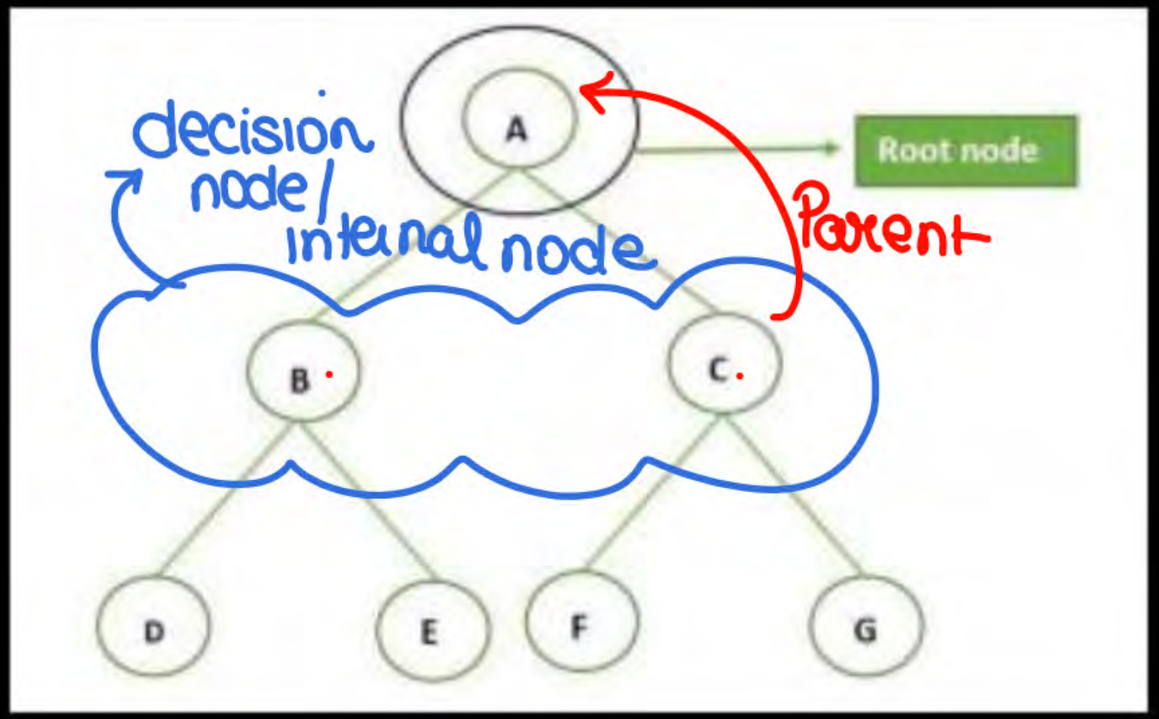
What is a Decision Tree

Some Important Terms

Importance of root node:

- The root node, which has no parents, is the highest node in the tree structure.
- The entire message is represented by this node, which is a global element.
- It may have one or more child nodes, but it is never recurring or able to have sibling nodes.

- Decision/Internal node are those which have children.
- Leaf Node are those which donot have any children.



- decision node ⇒ Those which have Parent and children both.
  - Here A is Parent of B, C and B, C are children of A
  - Root node ⇒ no Parent, but will have children

Decision Tree

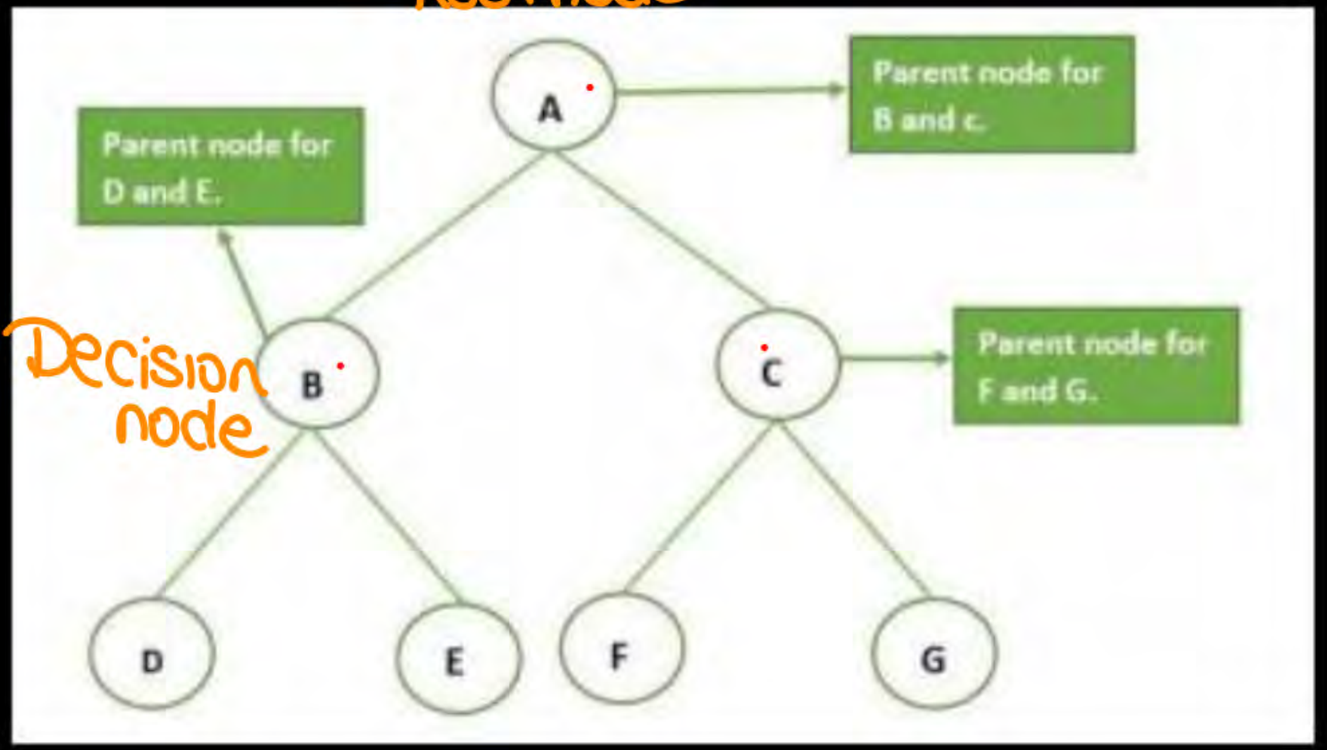
What is a Decision Tree

Some Important Terms

Parent Node

A is Parent of B, C
B " " " D, E
C " " F, G

Rootnode



Decision node

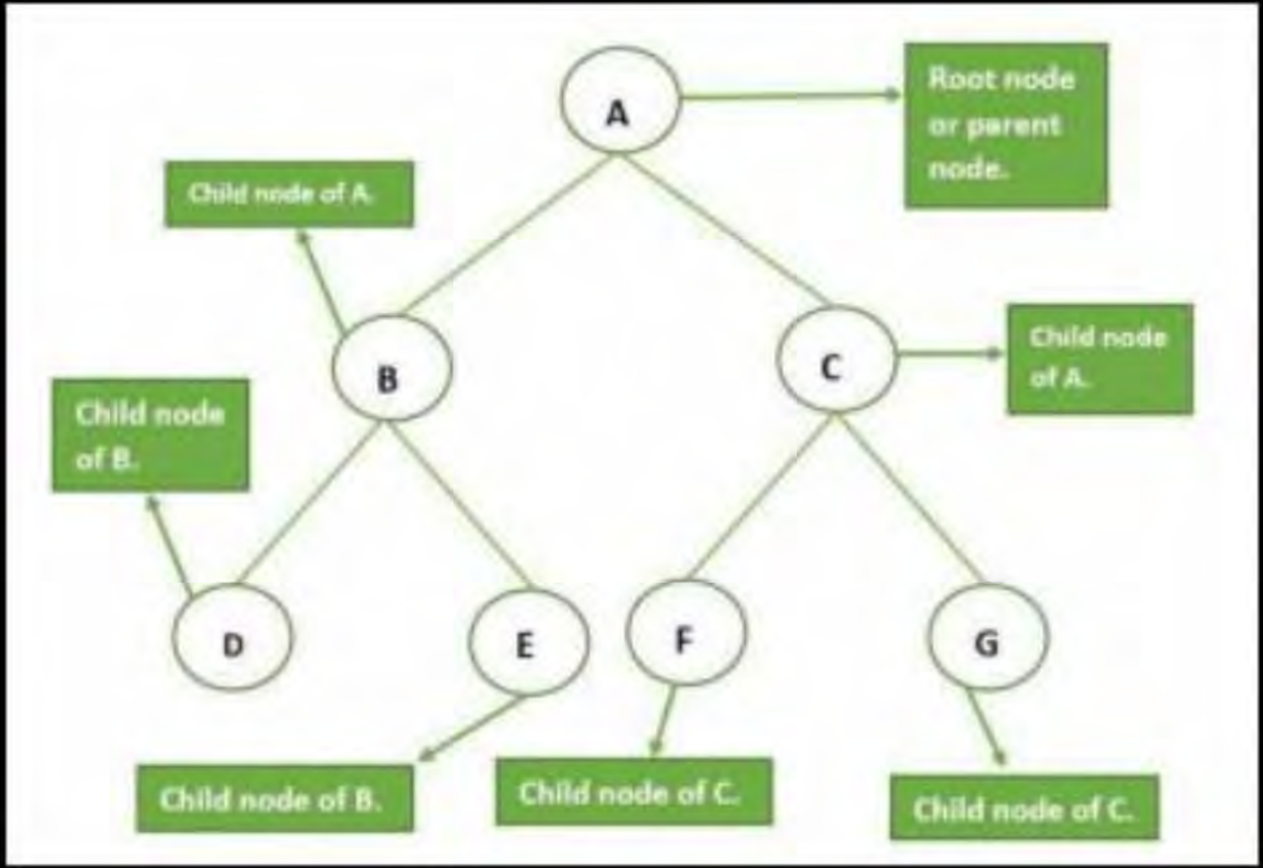
Decision Tree

What is a Decision Tree

Some Important Terms

Child Node

B, C are child of A
D, E " " " B
F, G " " " C



Decision Tree

What is a Decision Tree

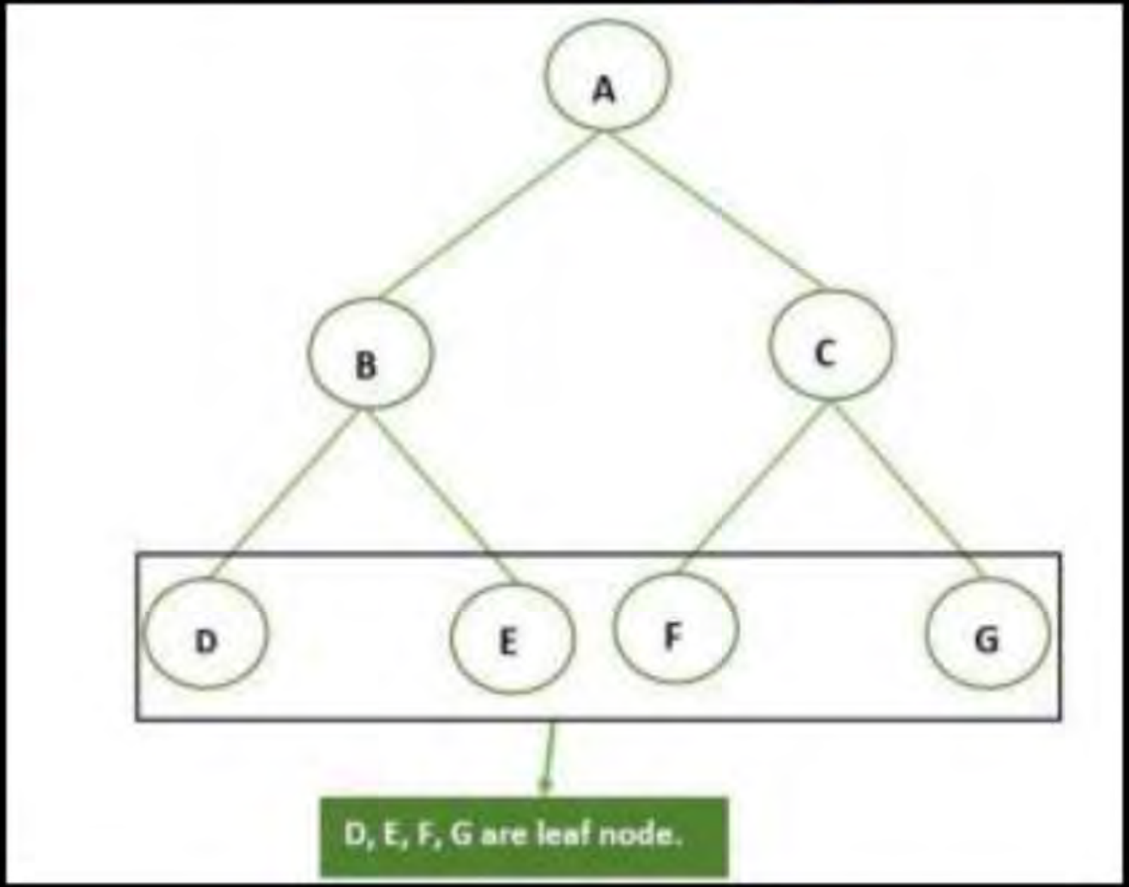
Some Important Terms

Leaf/Terminal Node

→ end node of Tree

→ These have no child but Parents.

→ While testing all decision is done @ the leaf node.



Decision Tree

What is a Decision Tree

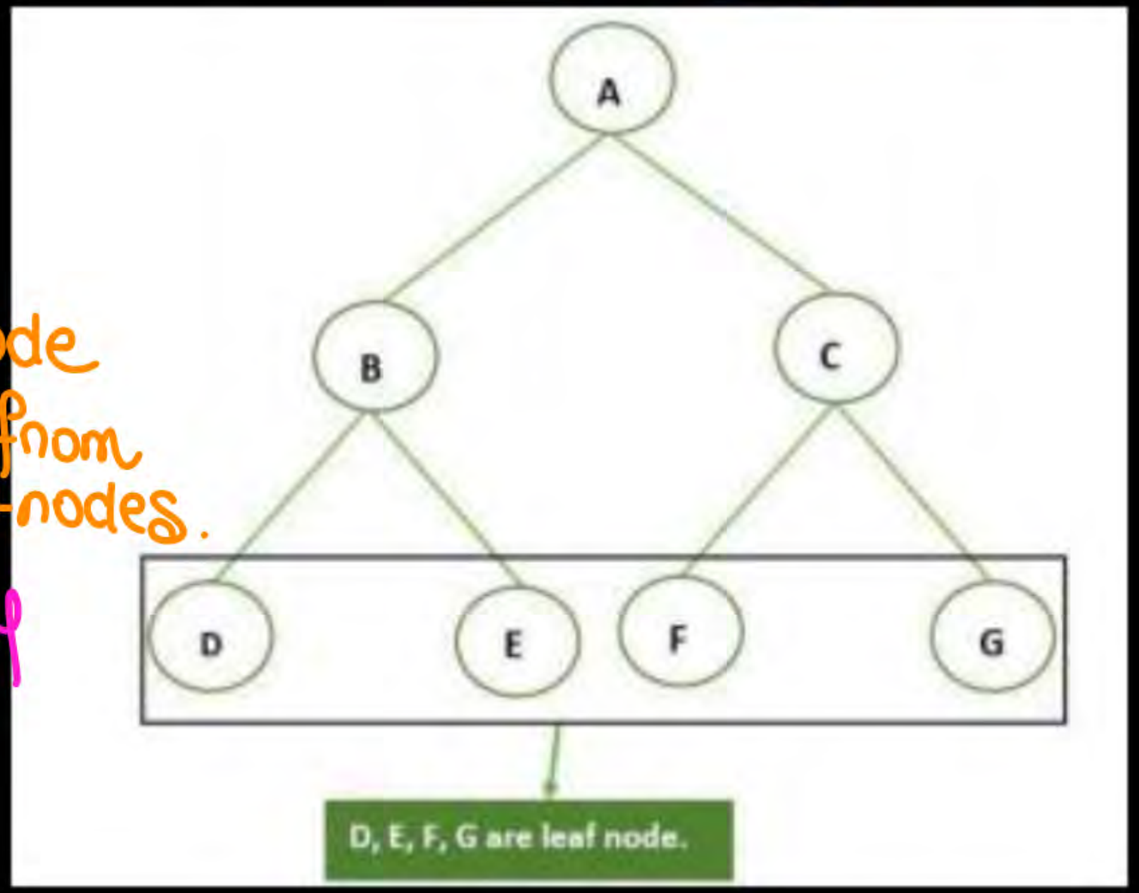
Some Important Terms

Height of the tree

The height of a tree is the total number of edges from a leaf node to any particular node usually a root node.

⇒ Height of any node ⇒ max no of node traversed from node to root nodes.

⇒ Height of tree ⇒ It's the height of deepest leaf node.



Decision Tree

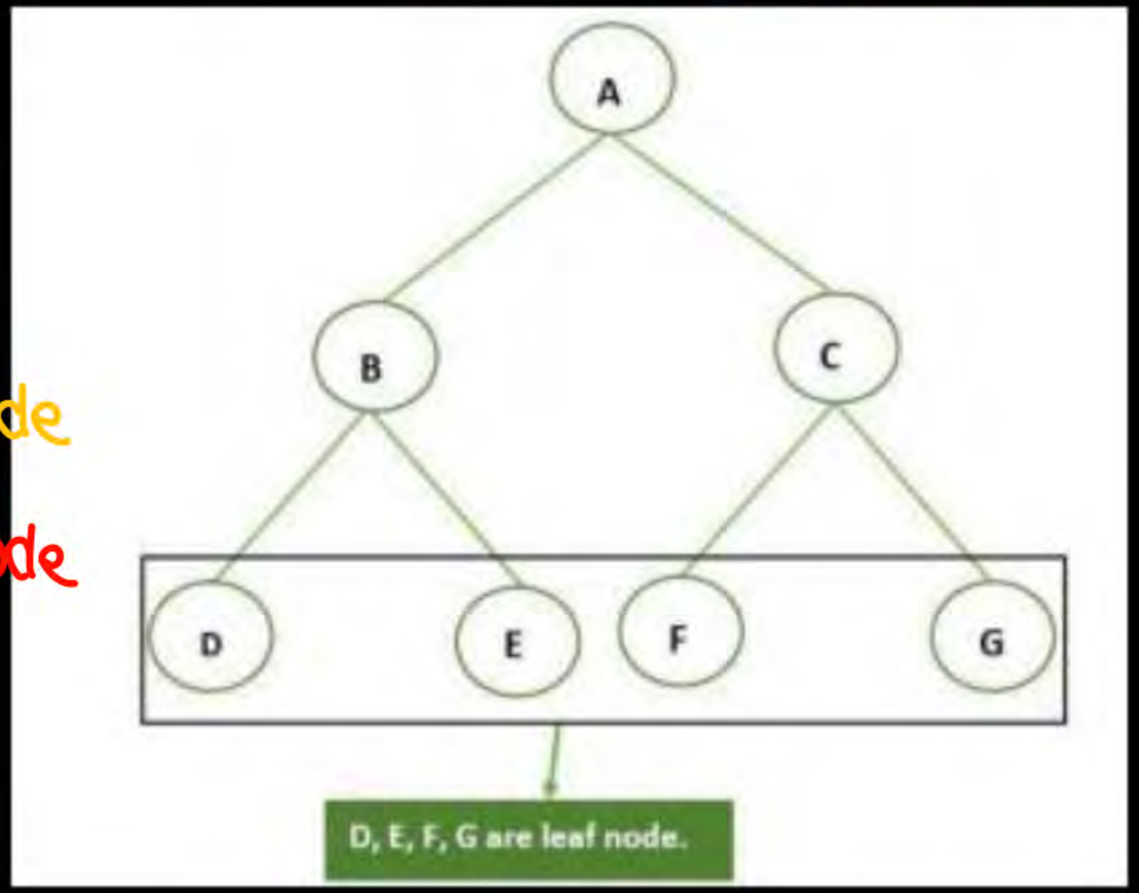
What is a Decision Tree

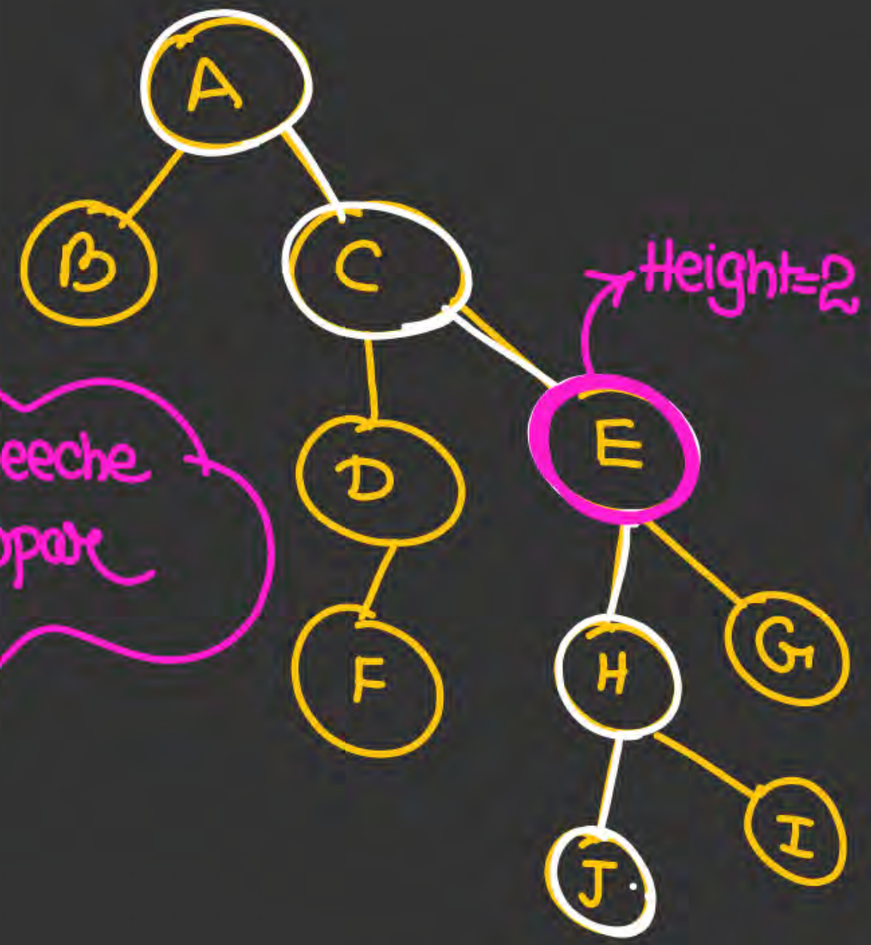
Some Important Terms

Depth of the tree

- depth of any node is the no of nodes b/w the node and leaf node

→ depth of tree ⇒ depth of root node





Height ⇒ neeche
depth ⇒ oopar

- Height of Tree ⇒ 4
- depth of tree ⇒ 4
- we can also count edges to get depth/height.

Decision Tree

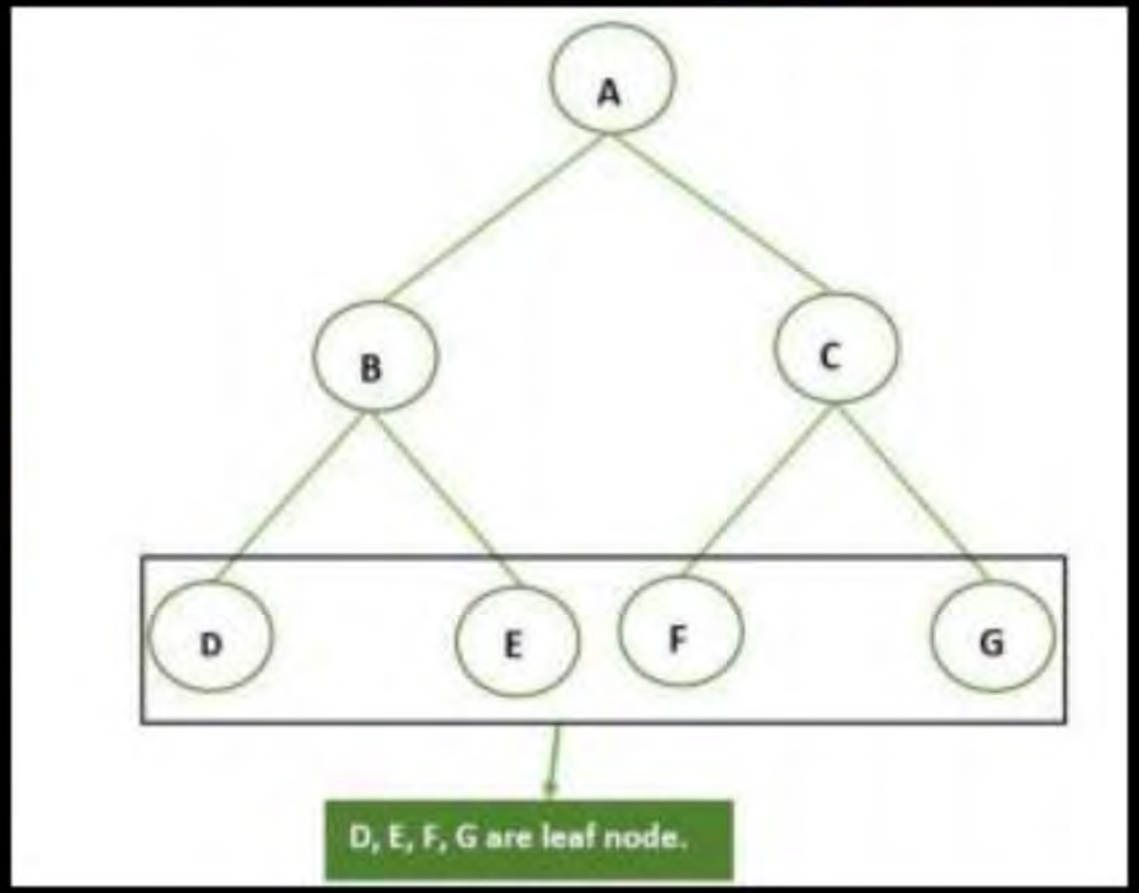
What is a Decision Tree

Some Important Terms

Degree of a Node in a tree

The Degree of a node in the tree is the summation of the number of child nodes connected to it, and the node which has the highest number of child nodes in a tree is also known as the Degree of a tree.

- degree of node ⇒ The number of Child node Connected



- degree of tree ⇒ max of degree of nodes.

Decision Tree

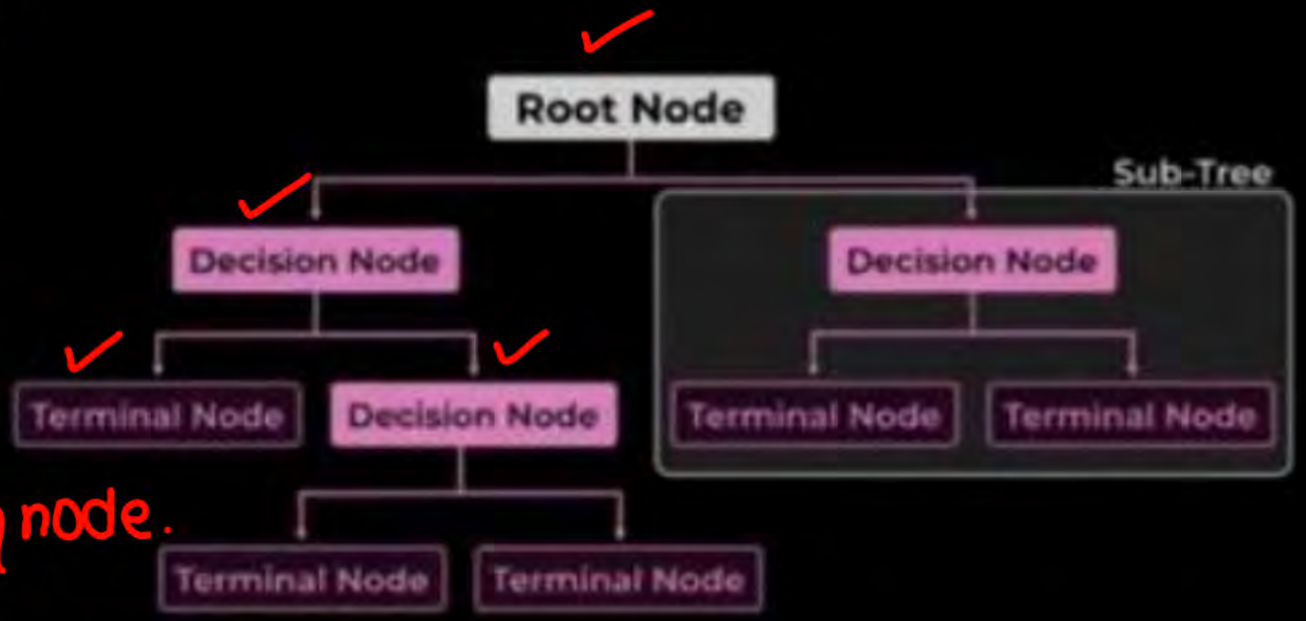
What is a Decision Tree

Some Important Terms

Internal / Decision Nodes

At each decision node we decide and divide input space into further regions.

So bifurcation/grouping of data is done @ Root node - decision node.



Decision Tree

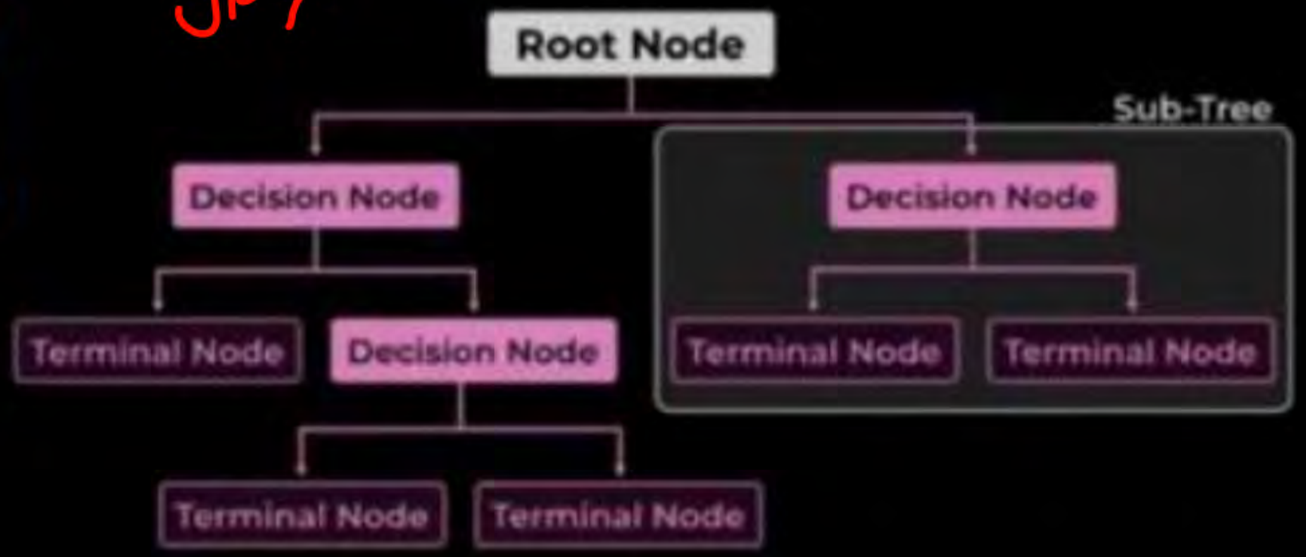
What is a Decision Tree

Some Important Terms
Decision Tree is Non Linear

Leaf / Terminal Nodes

→ finaldecision
$y_p \rightarrow$

The leaf node represent the label/ average value of the data point in case of classification and regression respectively.



THANK - YOU